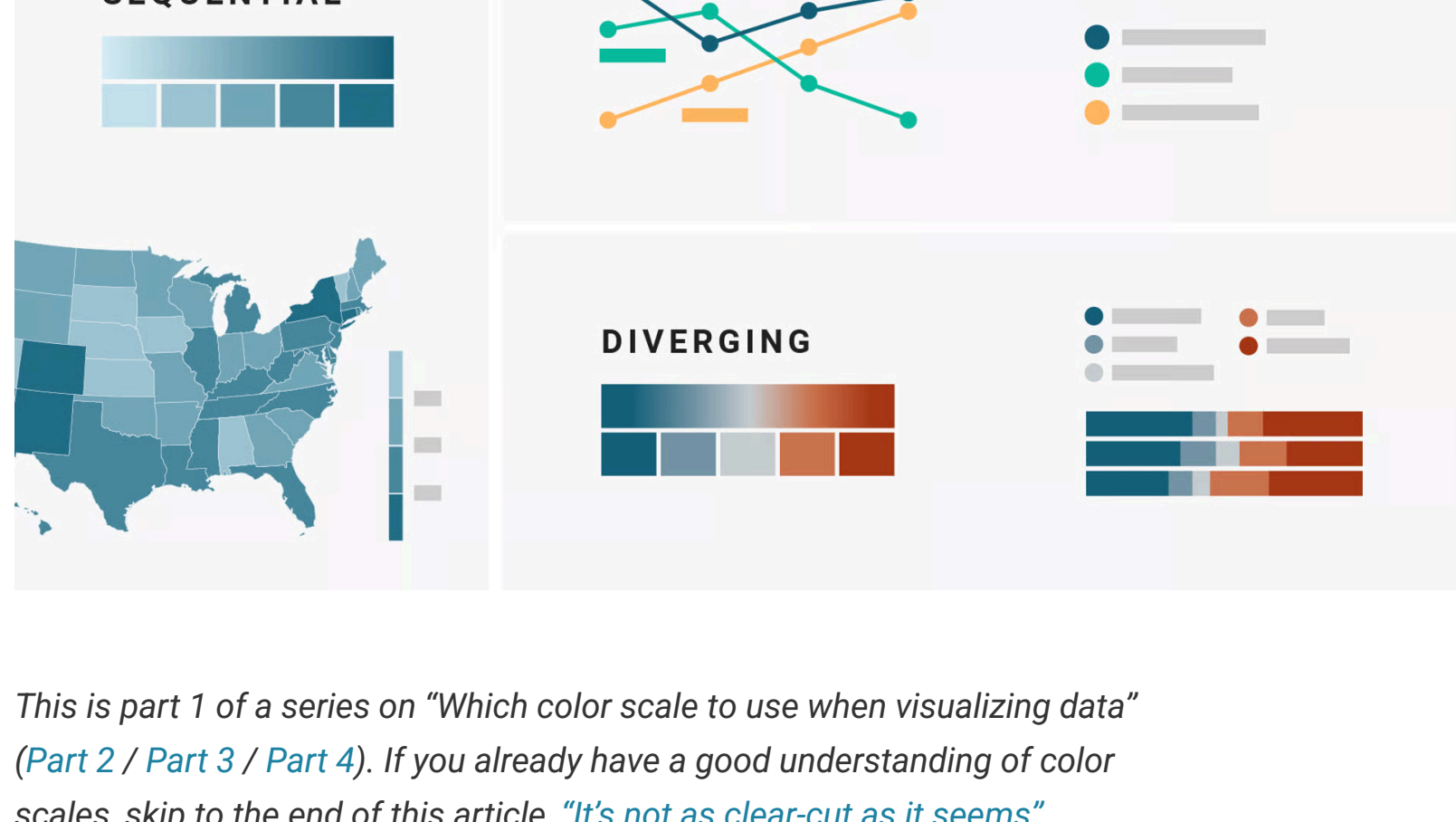


Color in data vis
March 16th, 2021

Which color scale to use when visualizing data

Lisa Charlotte Muth



This is part 1 of a series on "Which color scale to use when visualizing data" (Part 2 / Part 3 / Part 4). If you already have a good understanding of color scales, skip to the end of this article, "It's not as clear-cut as it seems".

When visualizing data, you're almost always working with color – e.g., with different hues (red, yellow, blue) for categories or color gradients (light blue, medium blue, dark blue) for maps.

If you use them to visualize data, hue palettes and gradients become "color scales." That's because they all "map" to some data. For example, every one of your hues stands for a certain category and every color in your gradient stands for a certain value (range).

This article gives you an overview of the different color scales. Let's start:

Index

- 1 What to color by
- 2 Hues: Categorical color scales
- 3 Color gradients in one direction: Sequential color scales
- 4 Color gradients in two directions: Diverging color scales
- 5 Highlighting or de-emphasizing
- 6 It's not as clear-cut as it seems

What to color by

Imagine you have a data set with the unemployment rate for each U.S. state for the last sixty years. What do you color by?

That depends on what type of visualization you're using (and which visualization type you need depends on what you want to communicate). Certain chart types nudge you to color by certain variables.

A choropleth map, for example, already encodes the states by position: Florida is in the bottom right, Texas in the bottom center, etc. There's no need to encode the states again with color. So you can encode the unemployment rate by color instead – for example, by making the fill color darker when the unemployment rate is higher.



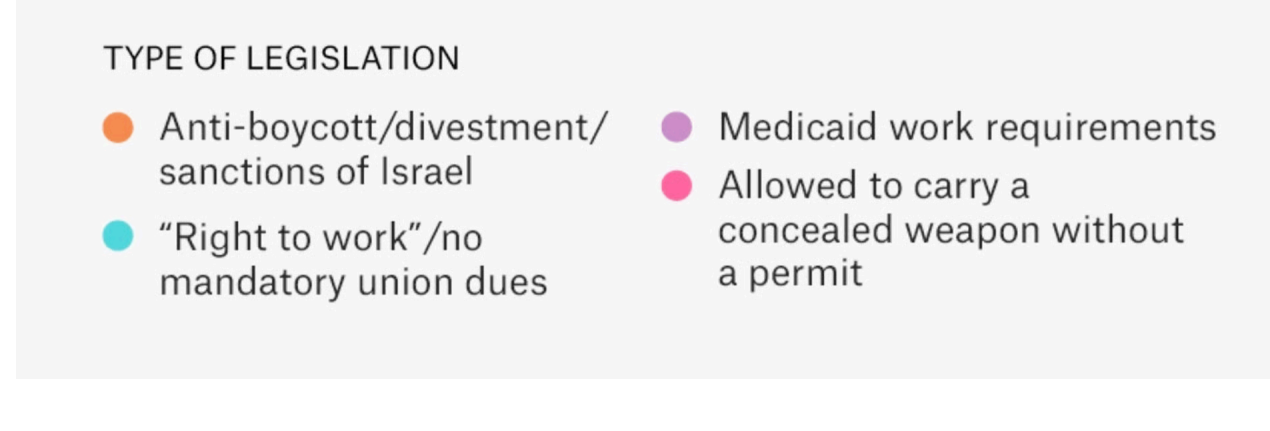
If you show the same data in a line chart, the opposite is the case. The unemployment rate is now encoded by position: The higher the line, the higher the unemployment rate. But to tell readers which line represents which state, you'll need colors.

So when you read about "encode categories with colors", "color quantitative values," etc. in the following minutes, then this applies to the part of your data set that you want to encode by color.

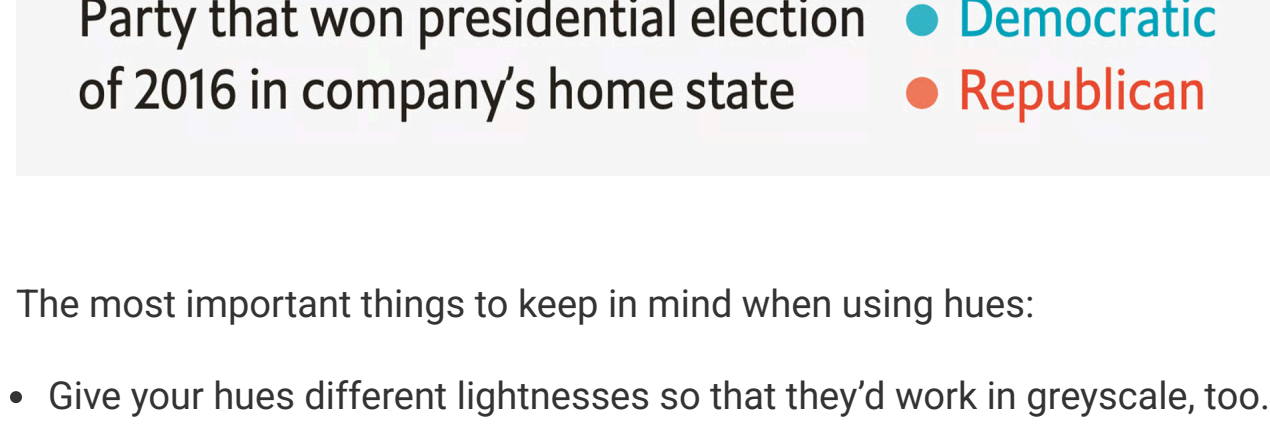
Hues

Categorical color scales ●●●●●

Hues are what a five year old would understand under "different colors": red, yellow, blue, etc. They're perfect to distinguish between categories that don't have an intrinsic order, like countries or ethnicities, genders or industries – that's why these categorical color scales are sometimes called *unordered* color scales. In such a color scale, colors say "I'm not worth more or less than these other colors here!"



Hues used in a legend for a chart by FiveThirtyEight



Hues used in a legend for a chart by The Economist

The most important things to keep in mind when using hues:

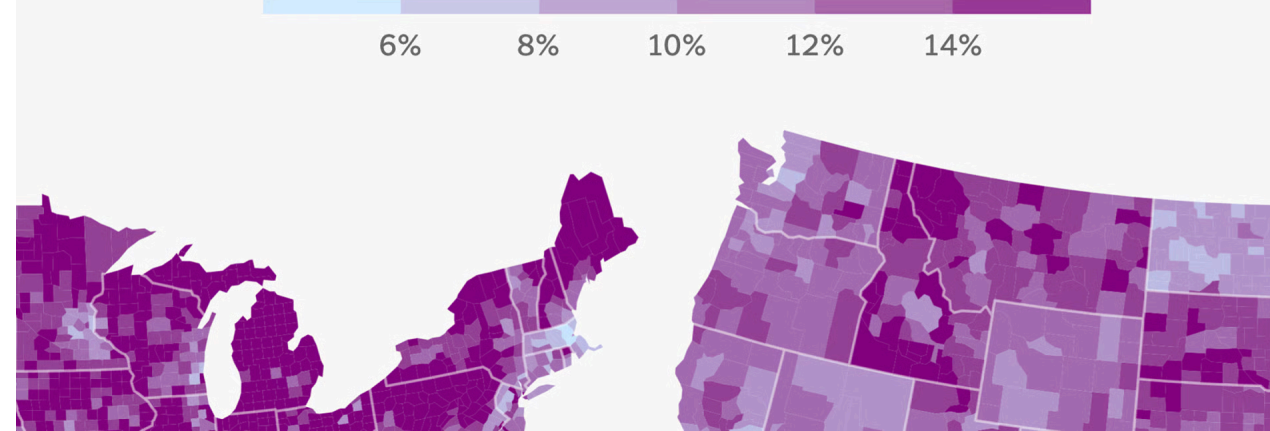
- Give your hues different lightnesses so that they'd work in greyscale, too. It makes them look better and easier to distinguish, which is especially important for colorblind readers. [Here are other tricks to make sure your hues are colorblind-friendly.](#)
- You don't need to design a color palette yourself. Lots of people have created amazing color combinations you can use. [Here's how to get inspired.](#) (If you want to design your own palette, [learn here how to make them beautiful.](#))

Color gradients in one direction

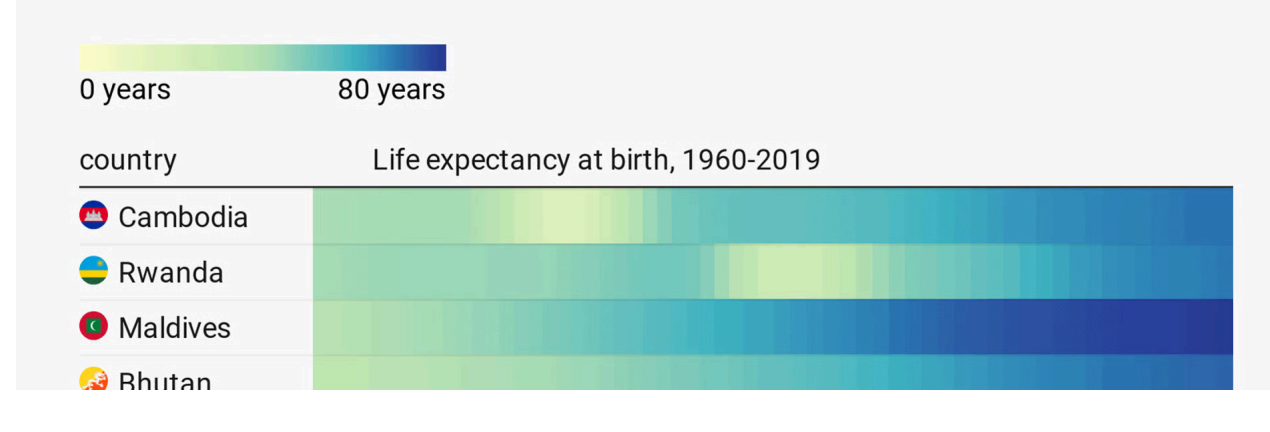
Sequential color scales ●●●●●

Sequential color scales are gradients that go from bright to dark or the other way round. They're great for visualizing numbers that go from low to high, like income, temperature, or age. A medium blue on a white background, for example, lets your readers know: "My value is a bit higher than the light blue and a bit lower than the dark blue."

Gradients can be classed (=split into brackets, also called *classified*, *stepped*, *quantized*, *graduated*, *binned* or *discrete*) or unclassed (=one continuous gradient):



Sequential, classed color scale, New York Times



Sequential, unclassified color scale, Datawrapper

The most important things to keep in mind when using sequential color scales:

- You can use only one hue in your sequential gradients (e.g., light blue to dark blue) – but almost all examples I show here use multiple hues (e.g., light yellow to dark blue). Using two or even more hues increases the color contrast between segments of your gradient, making it easier for readers to distinguish between them.
- To decide which data values correspond to which color in your gradient is called "interpolation" and has a massive influence on how readers perceive your values. [Here's how to choose the best interpolation for your data.](#)
- You can use the color gradients we offer in Datawrapper, or use tools like ColorBrewer 2.0 to copy some. [Here are tools to create color gradients yourself.](#)

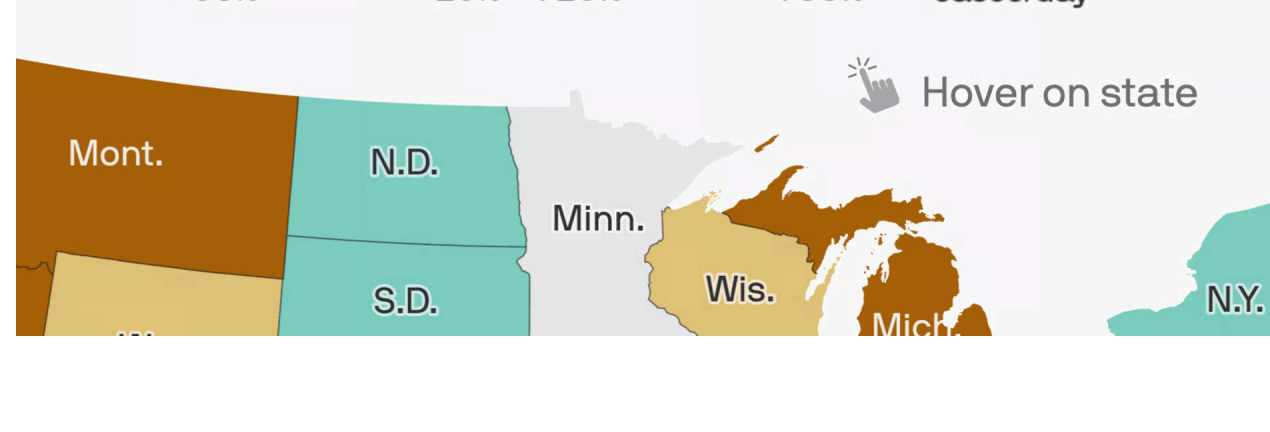
The same advice is true for diverging color scales:

Color gradients in two directions

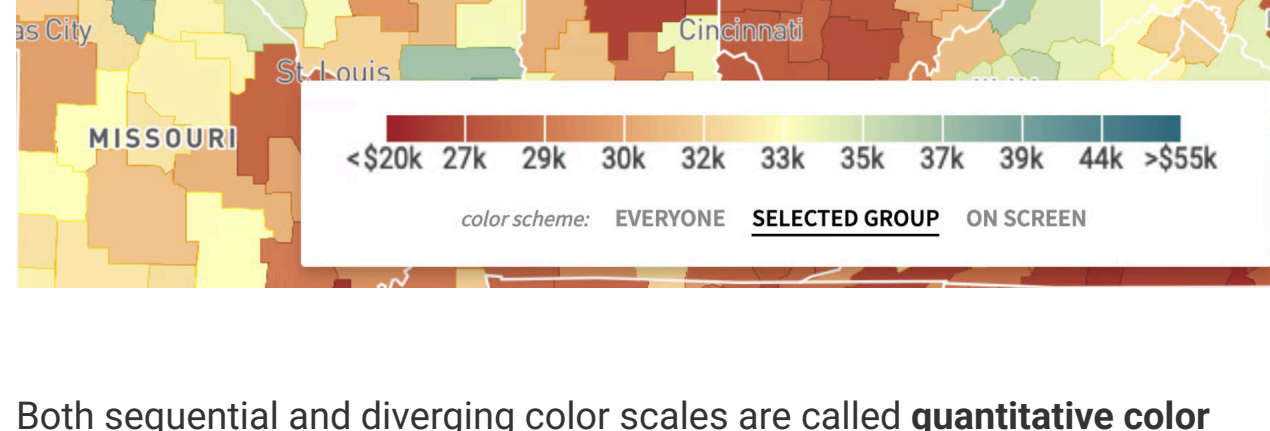
Diverging color scales ●●●●●

Diverging (also called *bipolar* or *double-ended*) color scales are the same as sequential color scales – but instead of just going from low to high, they have a bright middle value and then go darker to both ends of the scale in different hues. Diverging color scales are often used to visualize negative and positive values, election results, or Likert scales ("strongly agree, agree, neutral, disagree, strongly disagree").

Like sequential color scales, diverging ones can be classed or unclassified:



Diverging, classed color scale, Aol

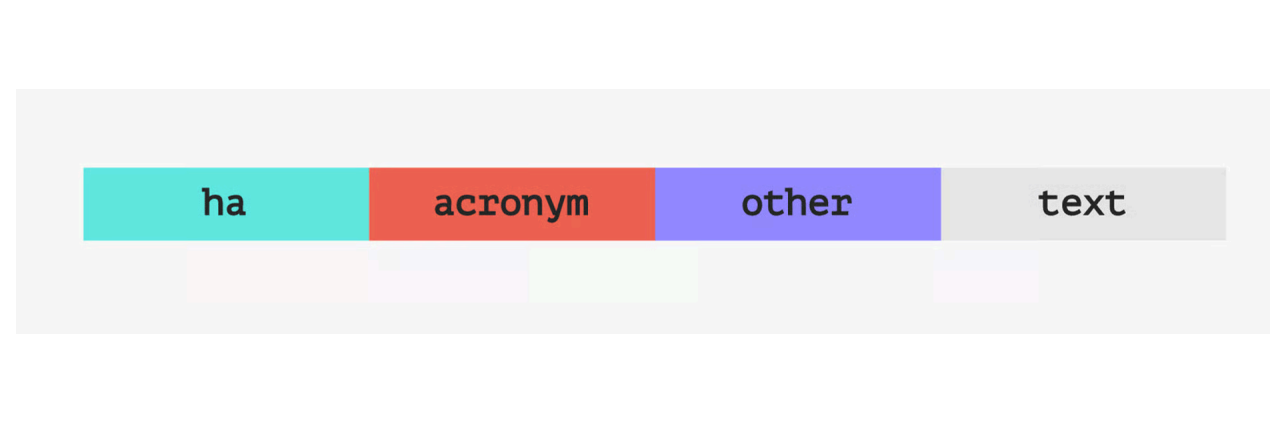


Diverging, unclassified color scale, Opportunity Atlas

Both sequential and diverging color scales are called **quantitative color scales**.

Highlighting/de-emphasizing ●●●●●

In any color scale, be it categorical, sequential, or diverging, you can highlight certain categories or value ranges that you consider especially important to your audience or story:



Unbinned gradient with a highlighted "0%" category, by The Guardian

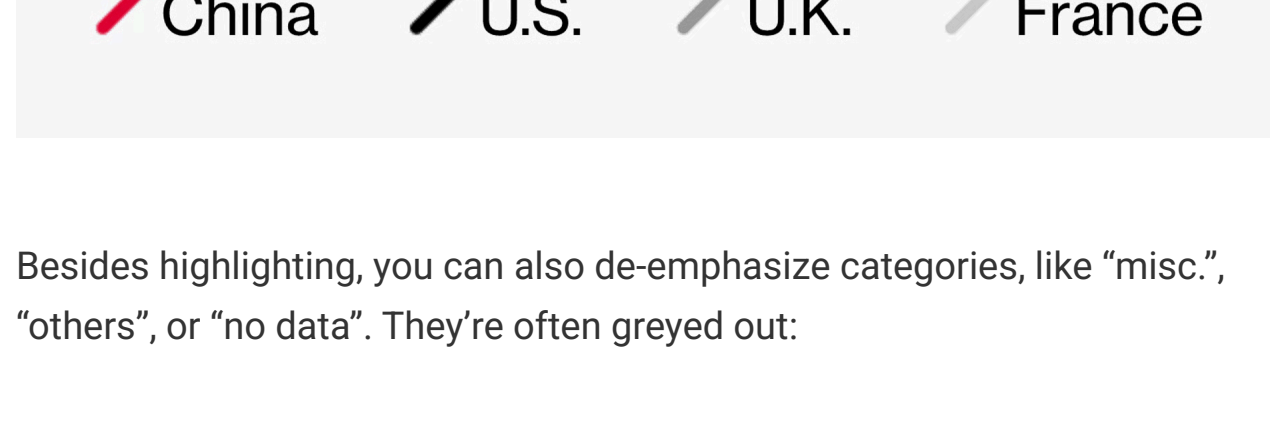


Chart by The Pudding with a de-emphasized "text" category

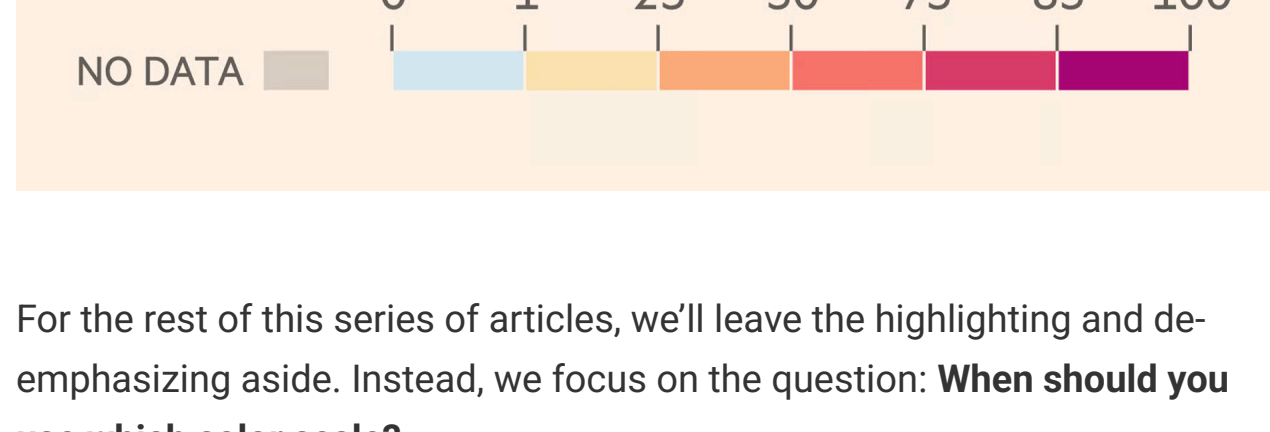
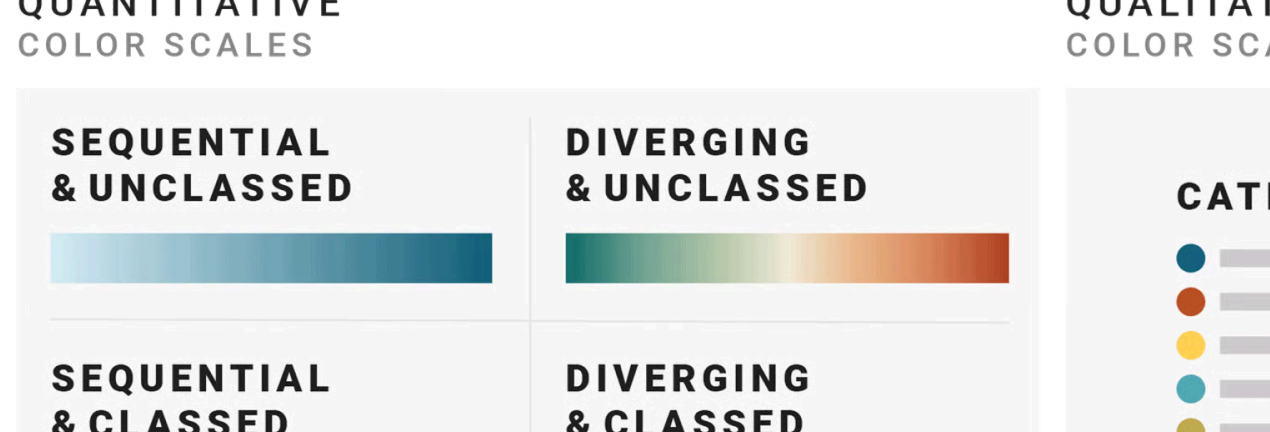


Chart by Bloomberg highlighting the "China" category

Besides highlighting, you can also de-emphasize categories, like "misc.", "others", or "no data". They're often greyed out:

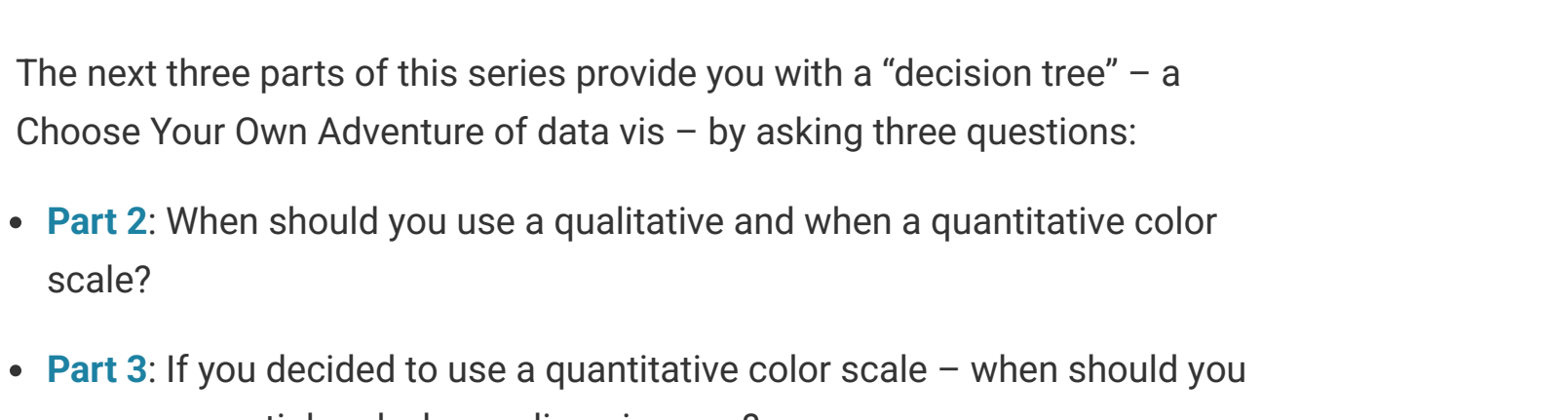


Binned gradient legend with extra "No data" category, by FiveThirtyEight

For the rest of this series of articles, we'll leave the highlighting and de-emphasizing aside. Instead, we focus on the question: **When should you use which color scale?**

It's not as clear-cut as it seems

The overview I gave you so far is what most data vis books will tell you. We can categorize these color scales like so:



But when looking at data visualizations, I noticed that the decision of which color scale to use is often not as obvious as many of these data vis books make us believe. Some data visualizations are using sequential color palettes, although they're visualizing categories. Or the same data is visualized with a diverging color scale in one publication and with a sequential one in the next. And sometimes with classed and other times with unclassified gradients.

What are the rules, the challenges, and the trade-offs?

Let's find out.

The next three parts of this series provide you with a "decision tree" – a Choose Your Own Adventure of data vis – by asking three questions:

- **Part 2:** When should you use a qualitative and when a quantitative color scale?
- **Part 3:** If you decided to use a quantitative color scale – when should you use a sequential and when a diverging one?
- **Part 4:** If you decided to use a quantitative color scale – when should you use a classed and when an unclassified one?

Let's start with the most basic question: *When should you use hues (a qualitative color scale) and when gradients (a quantitative color scale)?* [Click here to go to part 2.](#)

References

- Robert Simmon: **Subtleties of Color**, 2013. [Part 3: Different Data, Different Colors.](#)
- Claus O. Wilke: **Fundamentals of Data Visualization**, 2019: [Color scales](#)
- Adobe Spectrum: [Color for data visualization](#), 2019.
- Juuso Koponen, Jonatan Hildén: **The data visualization handbook**, 2019, p73-79.

Lisa Charlotte Muth (she/her, @lisacmuth, @lisacmuth@vis.social) is Datawrapper's head of communications. She writes about best practices in data visualization and thinks of new ways to excite you about charts and maps. Lisa lives in Berlin.

Liked this article? Maybe your friends will too:
[LinkedIn](#) [Bluesky](#)

COMMENTS

To post a comment you need to sign in via
[Twitter](#) or [GitHub](#) or [Mastodon](#)

NEWSLETTER

[Sign up to our newsletters](#) to get notified about everything new on our blog.

RELATED POSTS

- How to find & create good color palettes**
December 11th, 2024 by Lisa Charlotte Muth
- What to consider when choosing colors for race, ethnicity, and world regions**
October 9th, 2024 by Lisa Charlotte Muth
- Remind readers of the colors in your data visualization**
October 11th, 2023 by Lisa Charlotte Muth

ALL BLOG CATEGORIES

- Climate vis**
- Color book updates**
- Color in data vis**
- Covid vis**
- Data Vis Book Club**
- Data Vis Dispatch**
- Data vis do's & don'ts**
- Datawrapper News**
- Fix my chart**
- How to deal with data**
- Map Stories**
- Maps**
- Opinions**
- Team news and hiring**
- Uncategorized**
- Under the hood**
- Unwrapped**
- User stories**
- Weekly Charts**

[View a chronological list of our articles](#) [Learn more about our newsletters](#)

Product

- Charts
- Maps
- Tables
- Pricing

Feature highlights

- Custom themes
- Dark mode
- Print export
- Localization
- Accessibility
- Privacy
- Security

Solutions

- Media
- Finance
- Government
- Case studies

Resources

- Blog
- Academy
- FAQ
- River
- Webinars
- Training materials
- API Docs
- Get support

About

- About us
- Latest improvements
- Careers
- Privacy policy
- Terms of service
- Imprint
- Contact us